Assignment 3

Data packet Analysis using Wireshark

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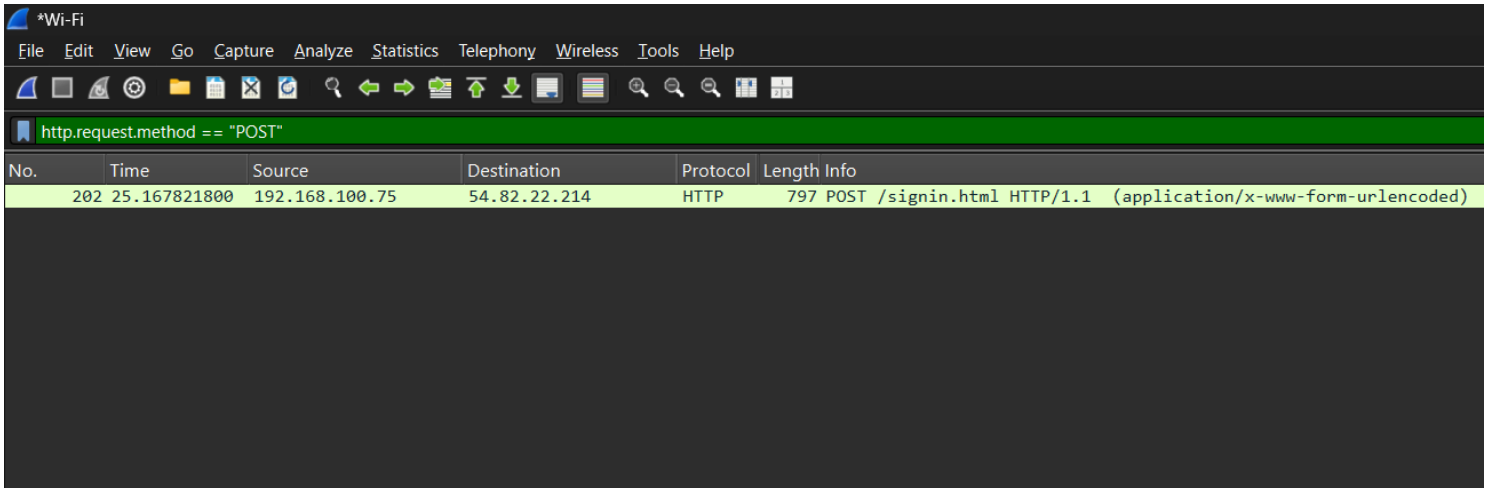
Tasks:

1. Install Wireshark
2. Search for HTTP website. Ensure that you are able to login by using default username and password.
3. Find the relevant IP Address of the website.
4. Apply relevant display filter to analyze the packet.
5. Analyze HTTP packet which will give you details of username and password.

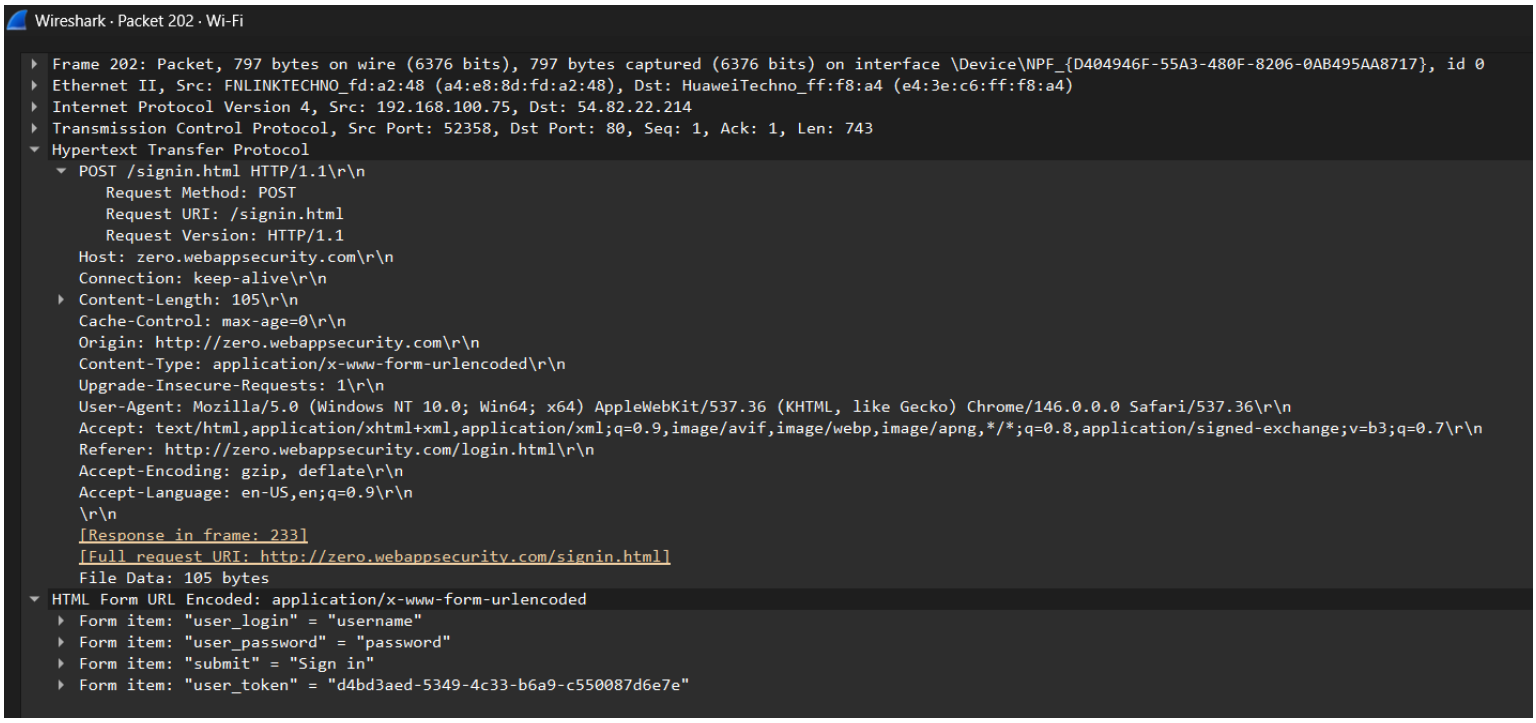
```
C:\Users\saimz>ping zero.webappsecurity.com

Pinging zero.webappsecurity.com [54.82.22.214] with 32 bytes of data:
Request timed out.
Request timed out.
Request timed out.
Request timed out.

Ping statistics for 54.82.22.214:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),
```

To identify the IP address of the target website, the ping command was executed in Command Prompt. The command used was ping zero.webappsecurity.com which successfully resolved the domain name and returned the IP address 54.82.22.214. This confirmed the actual network address of the web server. Although all four ping packets timed out with 100% packet loss, this is completely normal and expected because many web servers are configured to block ICMP ping requests for security purposes. The main goal of this step was still accomplished since the IP address 54.82.22.214 was successfully identified and noted down for later use as a display filter in Wireshark to isolate only the traffic related to this specific server.

188	25.164233000	54.82.22.214	192.168.100.75	TCP	66	80 → 52358 [SYN, ACK] Seq=0 Ack=1 Win=8192 Len=0 MSS=1412 WS=256 SACK_PERM
189	25.164597300	192.168.100.75	54.82.22.214	TCP	54	52358 → 80 [ACK] Seq=1 Ack=1 Win=65280 Len=0
190	25.164769300	54.82.22.214	192.168.100.75	TCP	66	80 → 57678 [SYN, ACK] Seq=0 Ack=1 Win=8192 Len=0 MSS=1412 WS=256 SACK_PERM
191	25.164769300	54.82.22.214	192.168.100.75	TCP	66	443 → 57669 [SYN, ACK] Seq=0 Ack=1 Win=8192 Len=0 MSS=1412 WS=256 SACK_PERM
192	25.164914500	192.168.100.75	54.82.22.214	TCP	54	57678 → 80 [ACK] Seq=1 Ack=1 Win=65280 Len=0
193	25.164958900	192.168.100.75	54.82.22.214	TCP	54	57669 → 443 [ACK] Seq=1 Ack=1 Win=65280 Len=0
194	25.165045100	192.168.100.75	142.251.37.142	QUIC	120	Handshake, DCID=e5d7dc89a3510a93
195	25.165184500	192.168.100.75	142.251.37.142	QUIC	73	Protected Payload (KP0), DCID=e5d7dc89a3510a93
196	25.165416500	192.168.100.75	142.250.187.42	QUIC	81	Handshake, DCID=fc5ab0de5dedbbab
197	25.166283000	192.168.100.75	142.250.187.42	QUIC	204	Protected Payload (KP0), DCID=fc5ab0de5dedbbab
198	25.166692400	192.168.100.75	142.251.37.142	QUIC	1093	Protected Payload (KP0), DCID=e5d7dc89a3510a93
199	25.167189100	192.168.100.75	54.82.22.214	TCP	1466	57669 → 443 [ACK] Seq=1 Ack=1 Win=65280 Len=1412 [TCP PDU reassembled in 200]
200	25.167189100	192.168.100.75	54.82.22.214	TLSv1	375	Client Hello (SNI=zero.webappsecurity.com)



After starting the packet capture on the Wi-Fi interface in Wireshark, the browser was opened and the login page at http://zero.webappsecurity.com/login.html was visited. The default credentials were entered into the login form and the Sign In button was clicked to submit them. Once the login request was sent, the Wireshark capture was stopped and a display filter was applied in the filter bar to narrow down the results. The filter used was http.request.method == "POST" which successfully filtered out all unnecessary traffic and displayed only the HTTP POST login request. Packet number 202 appeared in the results showing the source IP address as 192.168.100.75 which belonged to the local machine and the destination IP address as 54.82.22.214 which was the web server. The protocol was HTTP and the info column clearly showed POST /signin.html HTTP/1.1 with the content type application/x-www-form-urlencoded confirming that the login credentials were sent over an unencrypted HTTP connection.



Packet number 202 was selected and analyzed in detail through the Packet Details panel. The packet revealed a complete layer by layer breakdown of the network communication. The Frame layer showed 797 bytes were captured on the Wi-Fi interface. The Ethernet II layer displayed the source and destination MAC addresses. The Internet Protocol layer confirmed the source IP as 192.168.100.75 and destination IP as 54.82.22.214. The TCP layer showed

the connection was made from port 52358 to destination port 80 which is the standard HTTP port. The Hypertext Transfer Protocol layer showed the POST request directed to zero.webappsecurity.com. Most importantly the HTML Form URL Encoded section at the bottom revealed the login credentials in complete plain text. The field user_login showed the value "username" and the field user_password showed the value "password". This confirmed that HTTP transmits all form data including sensitive login credentials without any encryption making it possible for anyone on the same network to capture and read this information using Wireshark.